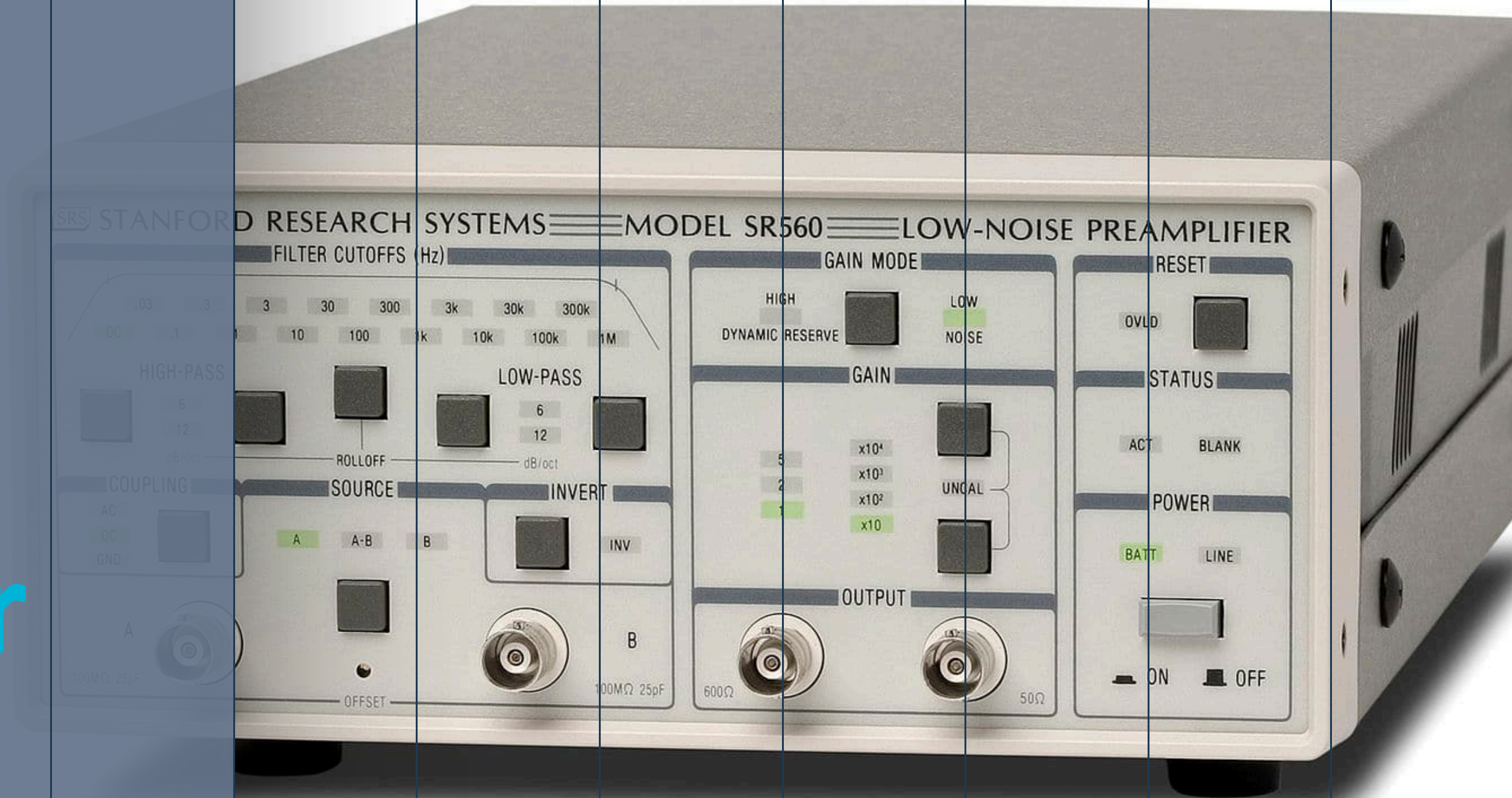


# Active Noise Canceling Pre-Amplifier



## The Interference Problem

- Mains hum: 50 Hz fundamental + harmonics (100, 150, 200 Hz...)
- Human vocal band: 300 Hz - 3.4 kHz (ITU-T G.711)
- Interference amplitude can exceed signal by >40 dB in unshielded environments
- Single-ended microphone inputs couple noise as a common-mode signal

## Why Analog Pre-Processing?

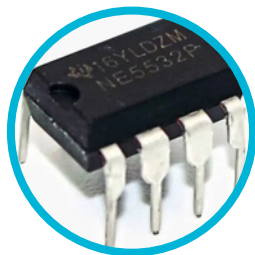
- ADC saturation: high-amplitude noise consumes dynamic range before digitization
- Digital DSP cannot recover clipped data — information is permanently lost
- Analog rejection must precede the ADC in the signal chain
- Differential input architecture rejects common-mode noise at the source

*Objective: Reject 50 Hz interference while preserving 300 Hz - 3.4 kHz vocal content with <1 dB passband ripple.*



Passive Components

- E24 series resistors
- COG/NPO ceramic capacitors (+-5%, temp-stable)
- Standardized values minimize deviation from calculated  $f_c$
- Sourced locally: Tronic.lk / Duino.lk
- Schottky diodes were used because their low forward voltage drop and fast switching characteristics
- Zener diodes were used to regulate and stabilize the output voltage

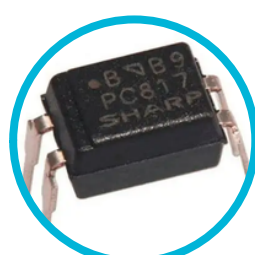


Op-Amp Selection

- Primary: NE5532 (low-noise, dual, audio-grade)
- Alternative: TL074 (quad, JFET input, low bias current)
- Both:  $V_{os} < 5\text{ mV}$ ,  $GBP \geq 10\text{ MHz}$
- Sourced locally: Tronic.lk / Duino.lk

FEATURES

- Small-signal bandwidth: 10MHz
- Output drive capability:  $600\Omega$ ,  $10V_{RMS}$
- Input noise voltage:  $5nV/\sqrt{Hz}$  (typical)
- DC voltage gain: 50000
- AC voltage gain: 2200 at 10kHz
- Power bandwidth: 140kHz
- Slew rate:  $9V/\mu s$
- Large supply voltage range:  $\pm 3$  to  $\pm 20V$
- Compensated for unity gain



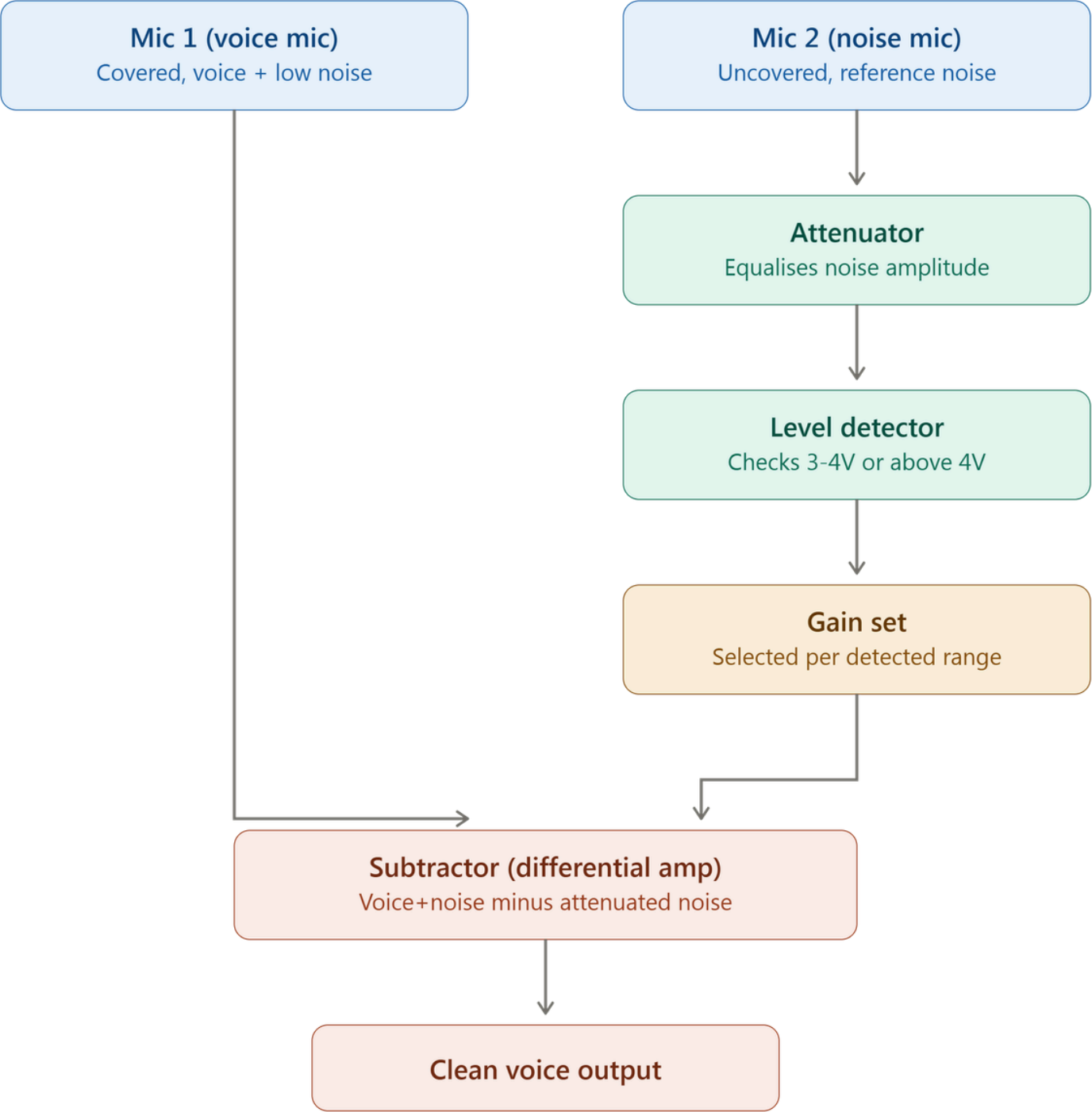
Optocoupler Selection

- An optocoupler was used to provide galvanic isolation between circuit sections, thereby reducing ground-loop hum, minimizing the coupling of external electrical noise into the low-level audio path, and protecting the sensitive preamplifier stage from interference and transients.

| Parameter                  |                                | Symbol    | Rating      | Unit  |
|----------------------------|--------------------------------|-----------|-------------|-------|
| Input                      | *1 Forward current             | $I_F$     | 50          | mA    |
|                            | *2 Peak forward current        | $I_{FM}$  | 1           | A     |
|                            | Reverse voltage                | $V_R$     | 6           | V     |
|                            | *1 Power dissipation           | P         | 70          | mW    |
| Output                     | Collector-emitter voltage      | $V_{CEO}$ | 35          | V     |
|                            | Emitter-collector voltage      | $V_{ECO}$ | 6           | V     |
|                            | Collector current              | $I_C$     | 50          | mA    |
|                            | *1 Collector power dissipation | $P_C$     | 150         | mW    |
| *1 Total power dissipation |                                | $P_{tot}$ | 200         | mW    |
| *3 Isolation voltage       |                                | $V_{iso}$ | 5           | kVrms |
| Operating temperature      |                                | $T_{opr}$ | -30 to +100 | °C    |
| Storage temperature        |                                | $T_{stg}$ | -55 to +125 | °C    |
| *4 Soldering temperature   |                                | $T_{sol}$ | 260         | °C    |



# 04 — Working Principle of the Circuit



- **Capture two inputs** – Mic 1 (covered) picks up voice + low-level noise; Mic 2 (uncovered) picks up reference noise only.
- **Equalise noise amplitude** – Attenuate Mic 2's noise signal so its level matches the noise leaking into Mic 1.
- **Detect noise level** – Measure the equalised noise amplitude and classify it: 3–4 V, or greater than 4 V.
- **Set gain** – Based on the detected range, adjust the attenuation/injection level using the power supply.
- **Combine signals** – Feed Mic 1's voice+noise signal and the conditioned noise reference into a subtractor (differential amplifier).
- **Cancel noise** – Subtract the attenuated noise from the voice+noise signal to produce a clean voice output.

## 05 — Signal-to-Noise Ratio (SNR) Analysis

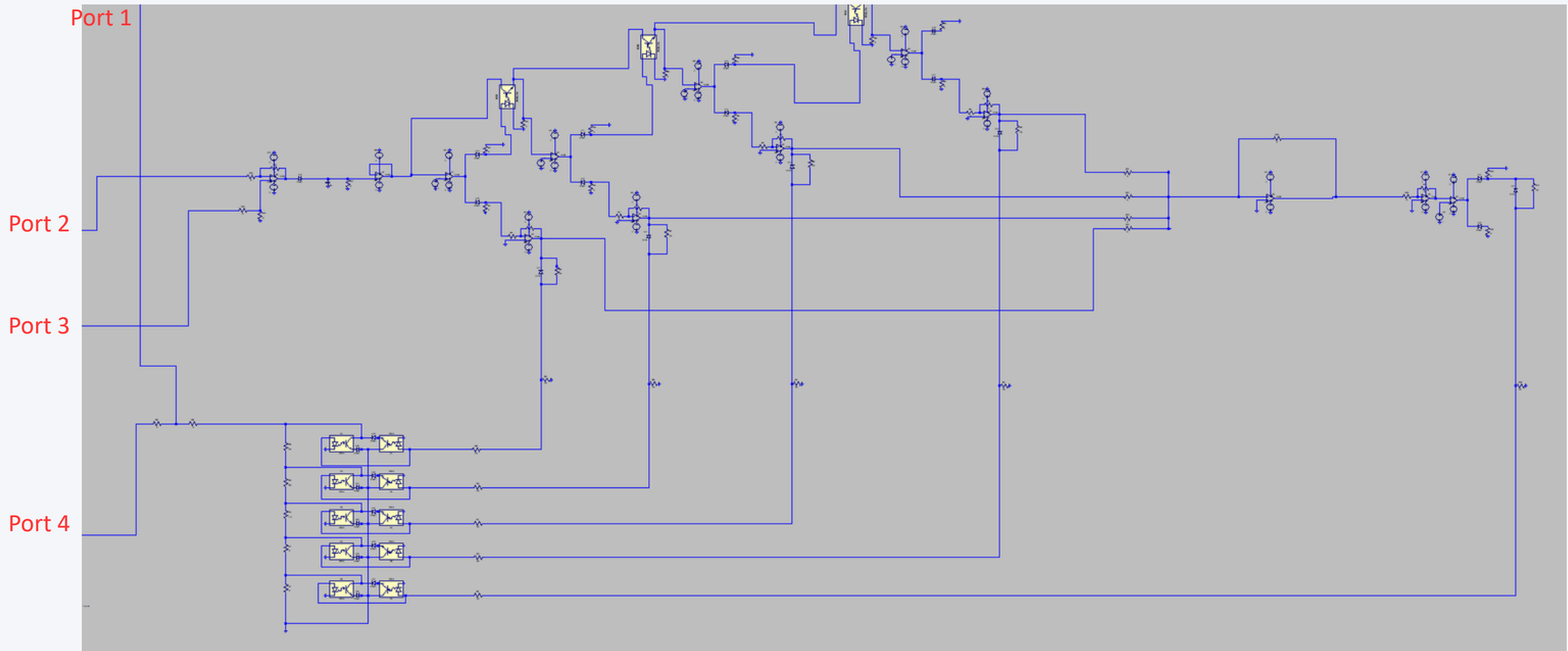
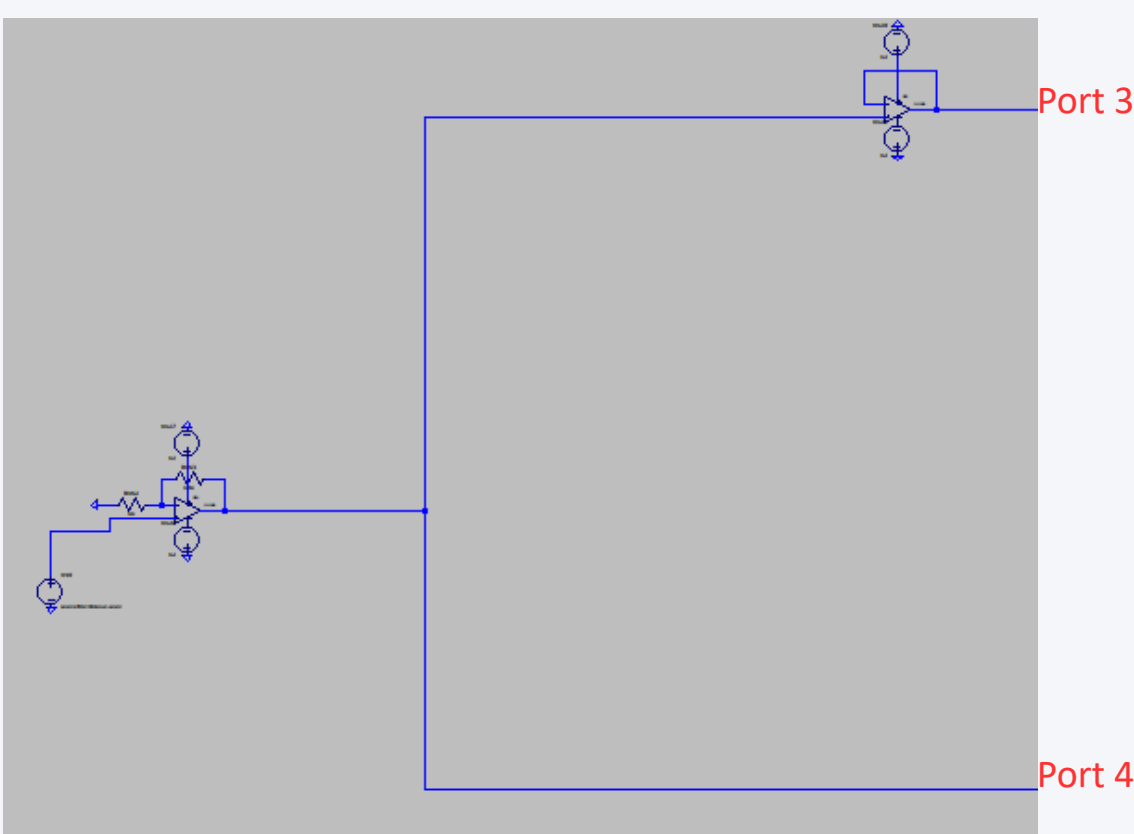
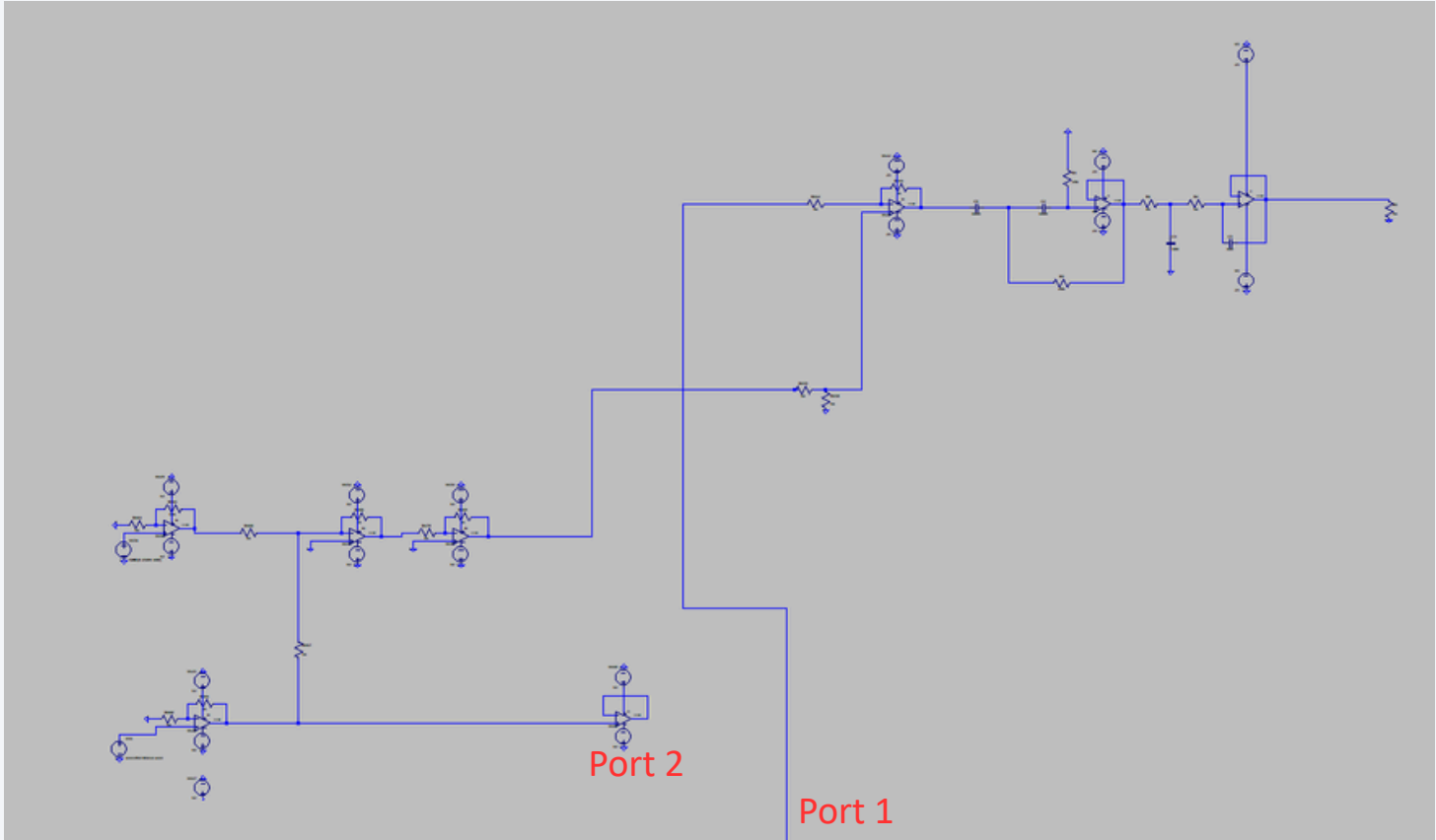
- SNR measures how strong the desired signal is compared to the unwanted noise.

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left[ \left( \frac{A_{\text{signal}}}{A_{\text{noise}}} \right)^2 \right] = 20 \log_{10} \left( \frac{A_{\text{signal}}}{A_{\text{noise}}} \right) = A_{\text{signal,dB}} - A_{\text{noise,dB}}.$$

- A higher Signal-to-Noise Ratio (SNR) indicates better audio quality because the desired signal is much stronger than the unwanted noise.
- A lower Signal-to-Noise Ratio (SNR) indicates poorer audio quality because the noise becomes more noticeable compared to the desired signal.

**SNR improvement = 5.329dB**

# 02 — Hardware Implementation





Thank You.....

